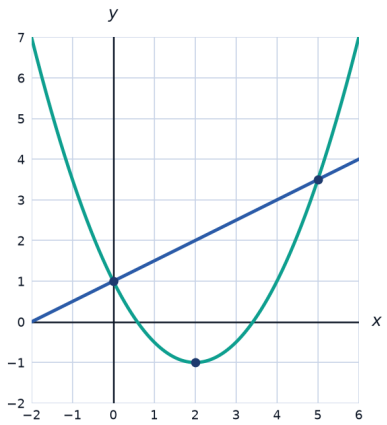
Lines & Angles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Deductive reasoning

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(5)$.

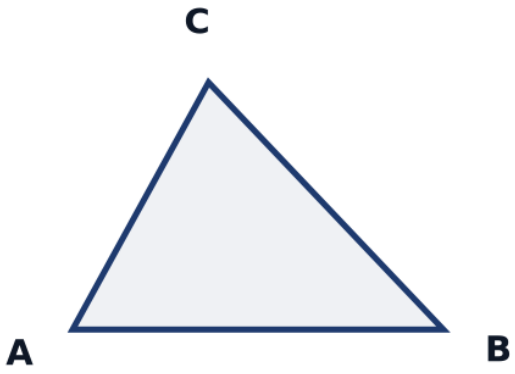


2 [Foundational] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

3 [Foundational] • Target: Properties of equality and congruence

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



4 [Foundational] • Target: Properties of equality and congruence

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

5 [Foundational] • Target: Properties of equality and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

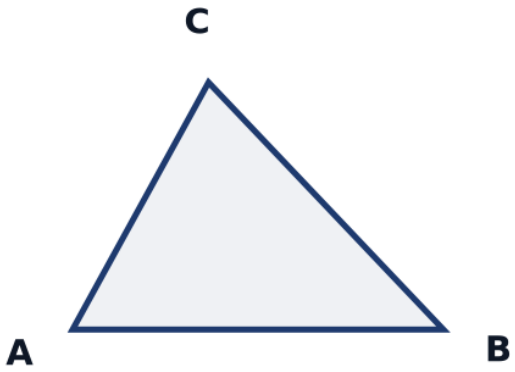
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6 [Foundational] • Target: Properties of equality and congruence

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Properties of equality and congruence

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Properties of equality and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

9 [Intermediate] • Target: Properties of equality and congruence

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

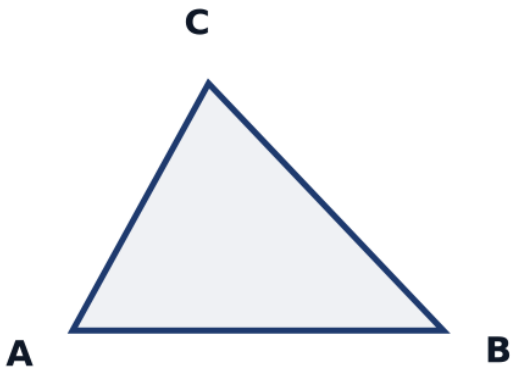
10 [Intermediate] • Target: Properties of equality and congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

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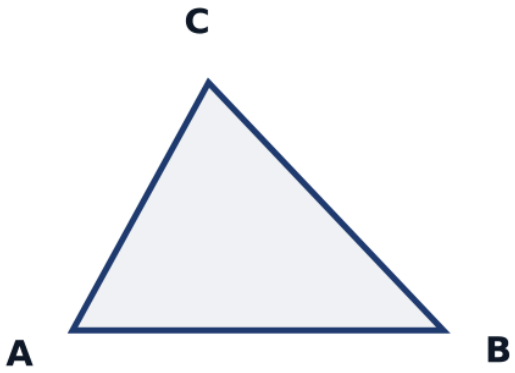
11 [Intermediate] • Target: Properties of equality and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



12 [Intermediate] • Target: Properties of equality and congruence

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: Properties of equality and congruence

A geometric figure scaled by factor $k = 5$ has an original side length of $8\ \text{cm}$. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: Properties of equality and congruence

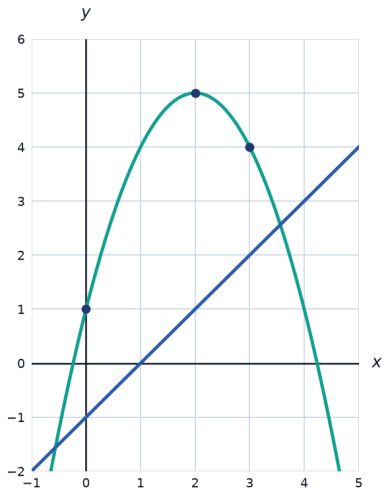
On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 7.01 Circles

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(2)$.



16 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 8.01 Area, Surface Area & Volume

A data set recorded for proving segment relationships consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

17 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 9.01 Transformations & Symmetry

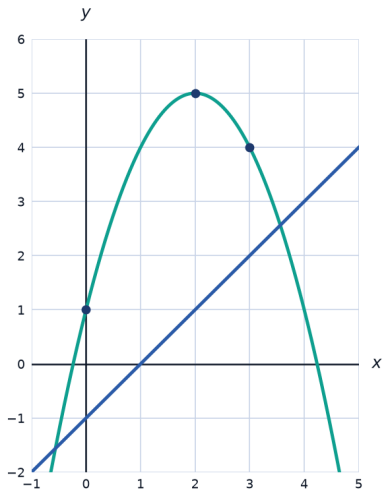
For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(3)$.

18 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 1.01 Foundations & Reasoning

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(4)$.

19 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 10.01 Coordinate Geometry

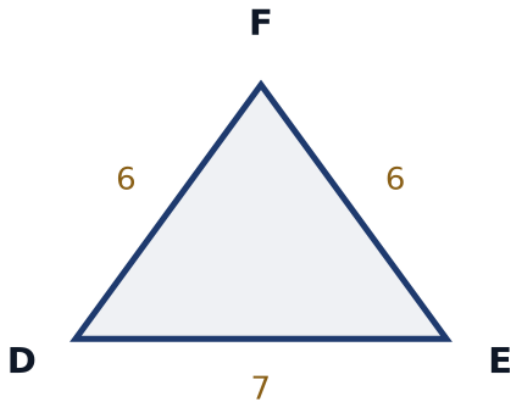
For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(5)$.



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20 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 11.01 Probability

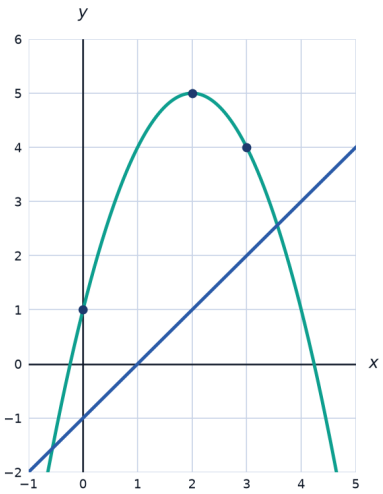
In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Conditional statements

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(2)$.



a. 15

b. 18

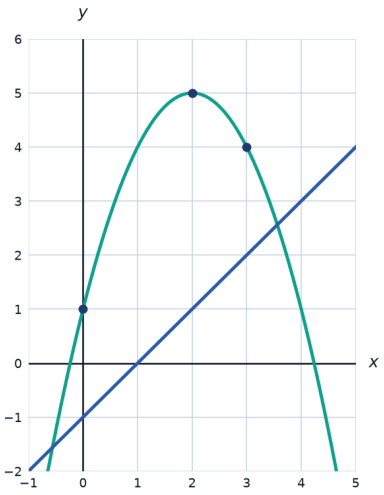
c. 6

d. 19

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22 [Foundational] • Target: Conditional statements

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

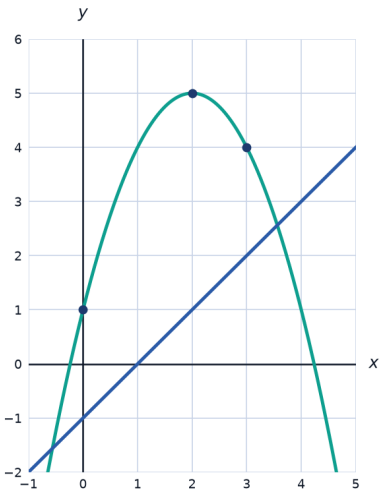
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

23 [Foundational] • Target: Conditional statements

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(5)$.



a. 24

b. 27

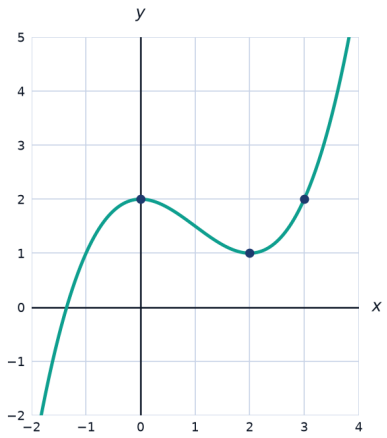
c. 15

d. 28

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24 [Foundational] • Target: Language of logic

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(2)$.



a. 11

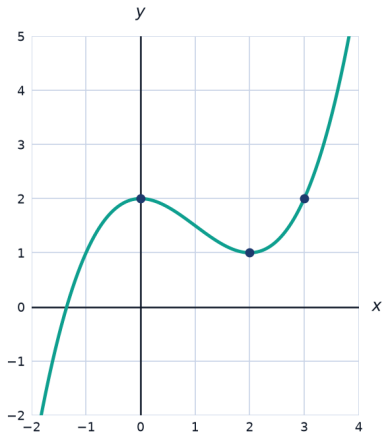
b. 15

c. 8

d. 15

25 [Foundational] • Target: Language of logic

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

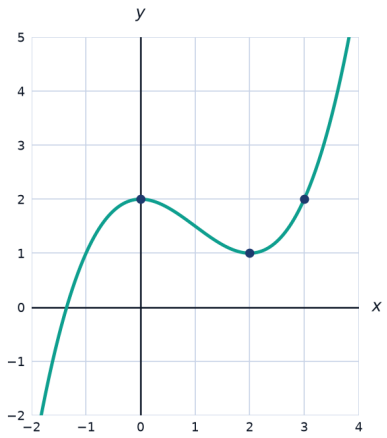
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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26 [Foundational] • Target: Language of logic

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(5)$.



a. 23

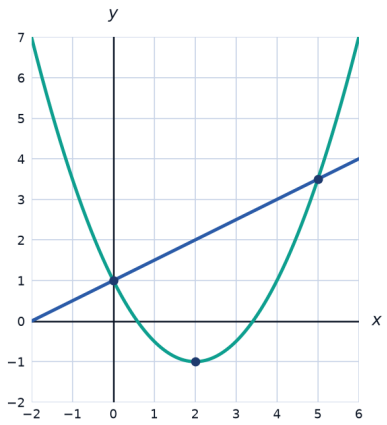
b. 27

c. 20

d. 27

27 [Foundational] • Target: Deductive reasoning

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(2)$.



a. 14

b. 19

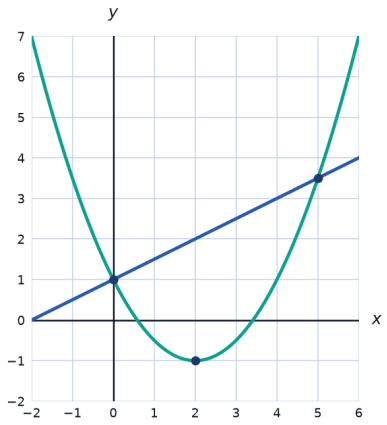
c. 10

d. 18

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28 [Foundational] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Conditional statements

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

30 [Intermediate] • Target: Conditional statements

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(3)$.

a. 18

b. 21

c. 9

d. 22

31 [Intermediate] • Target: Conditional statements

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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32 [Intermediate] • Target: Language of logic

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

33 [Intermediate] • Target: Language of logic

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(3)$.

a. 15

b. 19

c. 12

d. 19

34 [Intermediate] • Target: Language of logic

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

35 [Intermediate] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

36 [Intermediate] • Target: Deductive reasoning

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(3)$.

a. 19

b. 24

c. 15

d. 23

37 [Intermediate] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 10

c. 8

d. 11

39 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 10.01 Coordinate Geometry

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 8

b. 13

c. 11

d. 14

40 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

41 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 3.01 Triangles & Congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 14

b. 19

c. 17

d. 20

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42 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

43 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87°

b. 97°

c. 79°

d. 139°

44 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 80°

b. 90°

c. 72°

d. 139°

45 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 7.01 Circles

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

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